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1.0 Introduction

MindConnect is an innovative Artificial Intelligence (AI)-Based Diagnostic Support System designed for the preliminary detection and safe management of mental health conditions. The system employs a hybrid knowledge representation architecture to manage the complexity of psychological data, ensuring personalized and safe interventions.

The core functions rely on specific computational models:

- **Rule-Based Systems :** Used for preliminary diagnosis rules (2.1).
- **Frame-Based Systems :** Used for therapy selection by structuring user progress and severity data (2.2).
- **Semantic Networks :** Used for modeling user relationships (emotion, trigger, coping strategy) and risk assessment (2.3).
- **Predicate Logic :** Used for crisis triage by handling the uncertainty and degrees of truth in critical inputs (2.4).
- **Linguistic Patterns :** Used for language analysis to translate text into actionable facts about mood and issue dwelling (2.5).

Operating under a strict Ethical Rule against formal clinical diagnosis, MindConnect focuses solely on early detection, personalized support, and safe hand-offs.

2.0 Knowledge Representation

Knowledge Representation (KR) in AI refers to transforming complex, real-world knowledge like facts, rules, and relationships into a structured, formal language such as First-Order Logic (FOL) or network models that the AI can process logically. The goal of KR is to enable inference, allowing the system to derive new conclusions from existing information.

MindConnect employs KR schemes as the backbone of its reasoning capabilities. Specifically, the system uses a hybrid approach to handle the distinct types of psychological data it encounters:

- **Rule-Based Systems** define the **"what-if" rules** for diagnosis and crisis triage.
- **Frame-Based Systems** create the structured, digital **user profile** needed for personalized therapy selection.
- **Semantic Networks** map the **relationships** between concepts or emotions for contextual understanding and risk assessment.

By structuring raw input from Natural Language Processing (NLP) into these formal models, MindConnect ensures its decisions, whether a preliminary diagnosis, a change in therapy, or a safety check-in are consistent, logical, and safe.

2.1 First Order Logic (FOL) in MindConnect's for Preliminary Diagnosis

A Rule-Based System (RBS) is a knowledge representation scheme that uses a collection of IF-THEN statements to encode knowledge and perform logical reasoning. It is a foundational technology in artificial intelligence that has long been used in decision-making and problem-solving across various domains (GeeksforGeeks, 2025). MindConnect utilizes a RBS as the foundation for its AI-Based Diagnostic System. Its primary role is to ensure early detection of mental health conditions like Generalized Anxiety Disorder (GAD) and Mild-to-Moderate Depression. It rules map user-reported symptoms extracted via NLP from the conversation to potential conditions. Crucially, the RBS is constrained by the Ethical Rule that the AI will never attempt to give a formal clinical diagnosis. For safety constraints, it is designed only for support, detection, and safe hand-offs. Below is First-Order-Logic (FOL) for GAD and Mild Depression diagnosis.

Rule 1: Indicating Generalized Anxiety Disorder (GAD)

This rule formalizes the logic for indicating GAD based on persistent, excessive worry and an associated physical symptom (Mayo Clinic, n.d.).

$$\forall P\ (Symptom(P, ExcessiveWorry)\ \wedge\ Symptom(P,\ Restlessness)\ \wedge\ ReportsDuration(P, Weeks))\ \rightarrow Indicates(P, GAD)$$

Natural Language: For all persons (P), if P reports Excessive Worry AND this has lasted Weeks AND P reports Restlessness, THEN it indicates a potential Generalized Anxiety Disorder (GAD).

Rule 2: Indicating Mild-to-Moderate Depression

This rule formalizes a key diagnostic criterion for depression: the presence of a persistent low mood combined with a loss of pleasure (anhedonia) (Silva S., 2024).

$$\forall P\ (Symptom(P, LowMood)\ \wedge\ Symptom(P,\ Anhedonia)\ \wedge\ ReportsDuration(P, Weeks)) \rightarrow Indicates(P, MildDepression)$$

Natural Language: For all persons (P), if P reports Low Mood AND reports Anhedonia (loss of interest or pleasure) AND this has lasted Weeks, THEN it indicates a potential Mild Depression.

2.2 First-Order Logic (FOL) in MindConnect's for Therapy Selection

Frame-Based Systems can transform raw conversational input into structured knowledge that is essential for the logical reasoning engine by structuring a complex psychological information into discrete slots (e.g., Progress, Severity). Frame-Based Systems provide the structural architecture for this organization where it serves as the digital profile for each user, modeling their current and historical mental state (Minsky, 1974). By). This ensures the AI can quickly and consistently retrieve all necessary data points to select the safest and most effective therapeutic response.

Besides, a frame is ideally suited to model a single user's mental profile or a specific therapeutic intervention, hence, frame-based systems are critical to MindConnect because they bridge the gap between unstructured conversational data (what the user types) and structured, actionable knowledge required for intelligent decision-making (Minsky, 1974). This structure allows the MindConnect to efficiently achieve its goals of personalization, immediacy, and safety.

Rule 1: Intervention Selection Based on Stagnant Progress

This rule dictates a mandatory escalation of therapy when the user's current intervention is failing, focusing on the goal of continuous improvement.

$$\forall P:\ (HasProfile(P,\ Anxiety)\ \land\ Progress(P,\ Stagnant)\ \land\ (Severity\ (P)\ >\ 8)) \rightarrow\ Needs\ (P,\ NewExercise)$$

If any user (P) has an anxiety profile AND their progress is stagnant AND their severity score is above the threshold (8), THEN they need an elevated intervention.

Rule 2: Low Severity Intervention Default

This rule defines the conditions under which the system provides supportive, non-intensive check-ins, ensuring resources are appropriately managed for users who are stable or improving.

$$\forall P:\ (HasProfile(P,\ MildDepression)\ \land\ Progress(P,\ Improving)\ \lor\ (Severity\ (P)\ \leq\ 5)) \rightarrow\ Needs\ (P,\ DailyCheckin)$$

For any user (P), if they have a mild depression profile AND their progress is improving OR their severity score is low $(\leq\ 5)$, THEN they only require the standard, supportive $DailyCheckIn$ to maintain stability and monitor their state.

2.3 First-Order Logic (FOL) in MindConnect's for User Relationships

Semantic Networks enable MindConnect to organize psychological knowledge as interconnected nodes and relationships, allowing the system to represent concepts such as *User*, *Emotion*, *Trigger*, and *Coping Strategy* in a structured and relational manner. Instead of storing information as isolated data points, semantic networks map how these concepts influence or depend on one another—for example, *Stress* → *caused_by* → *Workload* or *Anxiety* → *alleviated_by* → *BreathingExercise*. This relational mapping helps the AI interpret user input within a broader psychological context, enabling it to recognize how multiple factors contribute to a user's mental state.

By modeling mental-health concepts as a network rather than a list, MindConnect can perform more advanced reasoning and inference. The system can trace emotional patterns, identify relevant interventions, and anticipate potential risks based on connected nodes. These structured relationships help the AI generate responses that are not only accurate but also contextually sensitive and consistent with established psychological principles. In doing so, semantic networks play a crucial role in supporting MindConnect's goals of delivering personalized, safe, and empathetic digital mental-health support.

Rule 1: Identifying Emotional Risk Through Negative Sentiment

This rule ensures the AI detects high-risk emotional states early to support safety and timely intervention.

Semantic Network Relationship

User → *HasEmotion* → NegativeSentiment → *HasIntensity* → HighIntensity → AtRisk

FOL Representation

$\forall$U,E (IsUser(U) $\wedge$ HasEmotion(U,E) $\wedge$ Negative(E) $\wedge$ Intensity(E,L) $\wedge$ High(L) → AssignRisk(U,AtRisk))

If a user consistently expresses negative emotions *AND* their emotional intensity is high, *THEN* the system must classify the user as "At-Risk" and trigger supportive actions.

Rule 2: Recommending Check-Ins When User Avoids Engagement

This rule maintains continuity in therapy by responding to patterns of disengagement.

Semantic Network Relationship

User → *HasEngagementLevel* → LowEngagement → *RelatedTo* → Avoidance → *TriggersAction* → CheckIn

FOL Representation

$\forall U\ ((IsUser(U) \wedge EngagementLevel(U,L) \wedge Low(L)) \vee AvoidsResponse(U) \rightarrow TriggerCheckIn(U))$

If a user has been inactive for an extended period *OR* avoids responding to reflective questions, *THEN* the system initiates a wellness check-in to maintain therapeutic connection.

2.4 First-Order Logic (FOL) in MindConnect's for Crisis Triage

The core job of the MindConnect system is to translate the user's various inputs (such as manual ratings, physiological data, or system metadata) into fuzzy inputs that the AI's rules can use. When processing these inputs, the system instantly analyzes them to determine degrees of mood, the severity of reported symptoms, and records the time of day. By converting these raw, imprecise measurements into degrees of membership in fuzzy sets, the system gives its fuzzy inference rules the input to work.

To achieve the goals of preliminary diagnosis and early detection, the MindConnect requires a strong way to represent human emotional and psychological states in a nuanced manner, a key approach when applying computational models to subjective areas like psychology (Deshpande, 2019). The system relies on its fuzzy rules to transform numerical inputs into actionable inferences for personalized responses and appropriate crisis triage that capture degrees of uncertainty, which is particularly useful in clinical decision support systems (Warren et al., 2000).

Rule 1: Identifying Emotional Distress

This rule triggers a responsive suggestion when a significant degree of negative self-rating and a high frequency of physical stress indicators are detected, suggesting a need for validation and structured therapeutic interaction.

Rule Syntax:

IF (User_Rating IS Poor) AND (Heart_Rate IS Elevated) THEN (Anxiety_Level IS High)

- The User_Rating input (e.g., a numerical value from 1 to 10) has a specific degree of membership in the fuzzy set Poor (e.g., $\mu=0.8$).
- The Heart_Rate input (e.g., bpm) has a specific degree of membership in the fuzzy set Elevated (e.g., $\mu=0.9$).
- The rule is evaluated using the minimum operator ($\min(0.8,0.9) = 0.8$), resulting in a firing strength that contributes to the final Anxiety_Level output.

Rule 2: Triggering Immediate Coping Action

This rule is designed for immediacy and crisis prevention, linking environmental context with emotional tone (e.g., anxiety) to deploy a specific and tailored coping mechanism.

Rule Syntax:

IF (Time_of_Day IS Night) AND (Anxiety_Level IS High) THEN (Offer_Intervention IS BreathingGuide)

- The Time_of_Day input (e.g., an hour value) has a specific degree of membership in the fuzzy set Night.
- The Anxiety_Level has a specific degree of membership in the fuzzy set High.
- The system uses the degrees of truth for both conditions to infer a weighted recommendation for a guided coping mechanism like a 2-minute breathing guide.

Rule 3: Crisis Triage

This rule handles crisis triage by escalating the response when indicators cross a critical threshold, requiring a high-level intervention.

Rule Syntax:

IF (Self_Harm_Indicator IS Present) OR (Suicide_Ideation IS Extreme) THEN (Triage_Level IS Emergency)

- The Self_Harm_Indicator has a high degree of membership in the fuzzy set Present based on specific inputs.
- The Suicide_Ideation input has a high degree of membership in the fuzzy set Extreme.The system uses the maximum operator for the OR condition (max($\mu_{Present}$, $\mu_{Extreme}$)).
- If the resulting degree of truth is high enough, the Triage_Level output is strongly associated with Emergency, which triggers an immediate alert to human crisis support or provides a list of emergency resources.

2.5 First-Order Logic (FOL) in MindConnect's for Language Analysis

The core job of Natural Language Processing (NLP) in MindConnect is to translate the user's feelings and thoughts into facts that the AI's rules can use. When sending the message, NLP instantly analyzed the text to determine the mood. This is to check if the user is stuck in thinking and records the time of day. By converting the complex language into simple facts, NLP gives the AI's logic rules the input to work.

To achieve the goals of preliminary diagnosis and early detection, the MindConnect requires a strong way to represent human emotional and psychological states. Given the reliance on Natural Language Processing (NLP) for analyzing user input (Glaz et al., 2020), these rules transform structured facts into actionable inferences for personalized responses.

Rule 1: Identifying Emotional Distress

This rule triggers an immediate, reflective response when negative language and repetitive dwelling are detected, suggesting a need for validation and structured therapeutic interaction.

$$\forall\, T\ (Mood\ (T, Down) \wedge StuckOnIssue(T) \rightarrow NeedsSupport(T))$$

The user's text in turn T expresses sadness, worry or any general unhappiness. The user's language pattern in turn T shows them endlessly repeating the same problem. The AI infers that the user is currently in emotional distress and requires an immediate therapeutic response.

Rule 2: Triggering Immediate Coping Action

This rule is designed for immediacy and crisis prevention, linking environmental context with emotional tone (e.g. anxiety) to deploy a specific and tailored coping mechanism (Horstkötter, 2025).

$$\forall\, U,E\ (Time\ (E, Night) \wedge Feels\ (E, Nervous) \rightarrow Offer(U, BreathingGuide))$$

The user's input event E occurred during non-standard hours (e.g. 10 pm to 6 am). The user's language tone in event E is classified as anxious, stressed or worried. The AI should immediately prompt the user U with a quick and guided coping mechanism like a 2-minutes breathing guide.

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